

(9b) WAP to check whether given graph is connected or not using DFS method.

```
#include <stdio.h>
```

```
#define MAX 10
```

```
int visited[MAX];
```

```
int adj[MAX][MAX];
```

```
int n;
```

```
void DFS(int v) {
```

```
    visited[v] = 1;
```

```
    printf("%d ", v);
```

```

for (int i=0; i<n; i++)
    if (adj[i][i] == 1 && !visited[i]) {
        DFS(i);
    }
}

int main() {
    int connected = 1;
    printf("Enter no. of vertices:");
    scanf("%d", &n);
    printf("Enter adjacency matrix: \n");
    for (int i=0; i<n; i++) {
        for (int j=0; j<n; j++) {
            scanf("%d", &adj[i][j]);
        }
    }
    for (int i=0; i<n; i++)
        visited[i] = 0;
    printf("DFS Traversal starting from vertex 0: \n");
    DFS(0);
    for (int i=0; i<n; i++) {
        if (!visited[i]) {
            connected = 0;
            break;
        }
    }
    if (connected) {
        printf("The graph is connected.");
    }
    else {
        printf("The graph is not connected.");
    }
    return 0;
}

```


Output:

Enter no of vertices: 5

Enter adjacency matrix:

0 1 0 1 0

1 0 1 0 1

0 1 0 0 0

1 0 0 0 1

0 1 0 1 0

DFS Traversal starting from vertex 0:

0 1 2 4 3

The graph is connected.